Mathematical Foundations of Political Methodology Lecture 2

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Recap/motivation

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- Logical reasoning is a skill obtainable via practice and study.
- Formal theory can remove ambiguity and imprecision, creating possibility for surprising discoveries: Arrow Impossibility Theorem.
- Today we will focus on demystification and vocabulary.

Today's plan

- Logical Notation
- Methods of Mathematical Proof
- Social Scientific Thinking

Proposition

Definition

A **proposition** *is a statement that is either true or false.*

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Three examples:

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Proposition: Oranges are a kind of vegetable.

Proposition: Superstition is the belief in the causal nexus.

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"not"	`!` (Bang)	$\sim$
"and"	`&`	$\cap$
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To understand compound propositions, we can use *truth tables*.

Truth Table

Truth tables were popularized by Ludwig Wittgenstein in his *Tractatus Logico-Philosophicus*

4.31 We can represent truth-possibilities by schemata of the following kind ('T' means 'true', 'F' means 'false'; the rows of 'T's' and 'F's' under the row of elementary propositions symbolize their truth-possibilities in a way that can easily be understood):

p	q	r
T	T	T
F	T	T
T	F	T
T	T	F
F	F	T
F	T	F
T	F	F
F	F	F

,

p	q
T	T
F	T
T	F
F	F

,

p
T
F

.

Proposition: Not(P)

The truth table for "Not(P)" is:

P	Not(P)
T	F
F	T

Proposition: P and Q

The truth table for "P and Q" is:

P	Q	P and Q
T	T	T
T	F	F
F	T	F
F	F	F

Proposition: P or Q

The truth table for "P or Q" is:

P	Q	P or Q
T	T	T
T	F	T
F	T	T
F	F	F

Proposition: Implication ($P \rightarrow Q$)

In an 'if then' statement, we say the antecedent P implies the consequent Q.

The truth table for "P implies Q" is:

P	Q	P implies Q
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In words:

- An implication is true exactly when the if-part is false or the then-part is true.
- In words: An implication is false exactly when the if-part is true and the then-part is false.

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- An implication is true exactly when the if-part is false or the then-part is true.
- In words: An implication is false exactly when the if-part is true and the then-part is false.
- "If my grandmother had wheels then she would have been a bike."

Alternative Intuition for Implication

The truth table for "not(P) or Q" is:

P	Q	not(P)	not(P) V Q
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

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Note that this is the same as implication.

Exercise: Pierce Law (1885)

$((P \rightarrow Q) \rightarrow P) \rightarrow P$

P	Q	$P \rightarrow Q$	$(P \rightarrow Q) \rightarrow P$	$((P \rightarrow Q) \rightarrow P) \rightarrow P$
T	T			
T	F			
F	T			
F	F			

Proposition: P IFF Q

The truth table for P IFF (if and only if) Q is:

P	Q	P IFF Q
T	T	T
T	F	F
F	T	F
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Example. This if-and-only-if statement is true for every real number x: $x^2 - 4 \geq 0$ iff $|x| \geq 2$

Example of IFF ($\iff$)

- Prove that: If A, B, and C are sets, then

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

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B.2) So $A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C)$.

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C.2) So $A \setminus (B \cap C) \supseteq (A \setminus B) \cup (A \setminus C)$.

D) So from B.2 and C.2, $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Proposition: Contrapositive

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Their truth table is:

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T	T	T	F	F	T
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F	T	T	F	T	T
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Example. "If I am hungry, then I am grumpy." is equivalent to "If I am not grumpy, then I am not hungry".

Proposition: Always vs. Sometimes True

Examples on "Always True":

Proposition: Always vs. Sometimes True

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For all $x \in \mathbb{R}$, $x^2 \geq 0$. Equivalent to write as: $\forall x \in \mathbb{R}$, $x^2 \geq 0$.

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Examples on "Sometimes True":

There exists an $x \in \mathbb{R}$ such that $5x^2 - 7 = 0$. Equivalent to write as: $\exists x \in \mathbb{R}$ such that $5x^2 - 7 = 0$.

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- Social Scientific Thinking

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Example of an axiom: There is a straight line segment between every pair of points.

Mathematical Proofs

Definition

A **mathematical proof** *of a proposition is a chain of logical deductions leading to the proposition from a base set of axioms.*

Classes of Results

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- Very important propositions are called **theorems**.
- A **lemma** is a preliminary proposition useful for proving later propositions.
- A **corollary** is a proposition that follows in just a few logical steps from a lemma or a theorem.

Methods for Mathematical Proofs

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If there is a skateboard on the sidewalk, my dog will bark at it.
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There is a skateboard on the sidewalk.
Therefore, my dog will bark.

Source of *modus ponens*

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and erroneously conclude: “Therefore, there is a skateboard on the sidewalk.”

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and erroneously conclude: “Therefore, there is a skateboard on the sidewalk.”

$$\frac{\mathrm{Q}, \quad \mathrm{P} \rightarrow \mathrm{Q}}{P \vee \sim P}$$

Source of affirming the consequent

P	Q	P implies Q
T	**T**	**T**
T	F	F
F	**T**	**T**
F	F	**T**

Methods for Mathematical Proofs: Direct Proof

So far we have shown that if we believe $P \rightarrow Q$, and P, Q follows.
Most political science claims are like $P \rightarrow Q$, so how do we prove this?
The most common form of proof is the *Direct Proof*

Methods for Mathematical Proofs: Direct Proof

So far we have shown that if we believe $P \rightarrow Q$, and P, Q follows.
Most political science claims are like $P \rightarrow Q$, so how do we prove this?
The most common form of proof is the *Direct Proof*

For this method, in order to prove that $P \rightarrow Q$, there are two steps: 1. Assume P is true; 2. Show that Q logically follows, usually from the definition of P.

Source of *Direct Proofs*

P	Q	P implies Q
T	**T**	T
T	F	F
F	T	T
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Methods of Mathematical Proof: Direct Proof

Theorem

If $0 \leq x \leq 2$, *then,* $-x^3 + 4x + 1 > 0$.

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Assume $0 \leq x \leq 2$. Then, x, $2 - x$, and $2 + x$ are all nonnegative.

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Assume $0 \leq x \leq 2$. Then, x, $2 - x$, and $2 + x$ are all nonnegative. Therefore, the product of these terms is also nonnegative. Adding 1 to this product gives a positive number, so:

$$x(2 - x)(2 + x) + 1 \geq 0$$

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Which is equivalent to:

$$-x^3 + 4x + 1 > 0$$

□

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Suppose $A = B$.

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To prove the contrapositive, there are two steps: 1. write, "We prove the contrapositive:", and then state $\sim Q \rightarrow \sim P$; 2. proceed as in the first method.

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Now, let's see an example.

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We prove the contrapositive: if $\sqrt{r}$ is rational, then, r is rational.
Assume that $\sqrt{r}$ is rational. Then, there exists integers a and b such that:

$$\sqrt{r} = \frac{a}{b}$$

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Since $a \times a$ and $b \times b$ are integers (integers are closed in multiplication):
r is also rational. □

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The step of proof by contradiction is: 1. Write, "We use proof by contradiction."; 2. Write, "Suppose P is false"; 3. Deduce something know to be false (such as a logical contradiction); 4. Write, "This is a contradiction. Therefore, P must be true."

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Suppose that there is a largest even integer, its name is k.
Since k is even, it has the form $2n$, where n is also an integer. Integers are closed under addition. Consider $k+2$. $k+2=(2n)+2=2(n+1)$.

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So $k + 2$ is even.

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So $k+2$ is even.
But $k+2$ is larger than k. This contradicts our assumption that k is the largest even integer $\square$. $\square$

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$\Leftarrow$ Suppose $A \subset B$ and that $B \subset A$.

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$\Leftarrow$ Suppose $A \subset B$ and that $B \subset A$. Now, by way of contradiction, suppose that $A \neq B$.
$A \neq B$ only if there is $x \in A$ and $x \notin B$ or a $y \in B$ and $y \notin A$.

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$\Leftarrow$ Suppose $A \subset B$ and that $B \subset A$. Now, by way of contradiction, suppose that $A \neq B$. $A \neq B$ only if there is $x \in A$ and $x \notin B$ or a $y \in B$ and $y \notin A$. But then, either $A \not\subset B$ or $B \not\subset A$, contradicting our initial assumption. $\square$

An aside on Hypothesis Testing.

We must decide what to reject upon arriving at a contradiction. A related point occurs in hypothesis testing.

- We suppose some ‘null hypothesis’ is true: there is no effect of pasta on weight gain.
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- We suppose some ‘null hypothesis’ is true: there is no effect of pasta on weight gain.
- We run an experiment.
- We find evidence that is unlikely to arise by chance if pasta does not affect weight gain and our experiment was well run.
- Do we conclude that pasta causes weight gain or that our experiment was not well run?

END VIDEO

Proof by Induction.

We want to show that a property holds for all n, where n is an integer.

1. Prove it holds for $n = 0$ (or $n = 1$).
2. *Assume* it holds for arbitrary n.
3. Prove it must hold for $n + 1$.

Example of proof by Induction.

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   - Thus, $|2^Y| = 2^n + 2^n = 2 * 2^n = 2^{n+1}$.
4. This proves our result. $\square$

- Proposition
- Methods of Mathematical Proof
- Social Scientific Thinking

Social Scientific Thinking

A *positive theory* in social science asserts that two or more concepts or events are related to each other and specifies the reason why.
Reasons why are called ‘mechanisms’.

We may aim to explain (interpret) or predict.

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This is usually a question of research design and logical merit.

Example. If the experiment engages in proper randomization, it may have internal validity.

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Example. If the randomized controlled trial is conducted on a representative sample, it may have external validity.
We may be concerned that our study of war from 1853-1856 doesn't travel to 264 BC.

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Example. A hypothesis can be: economic fluctuation causes civil conflict.

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Example. The rational choice model of political candidates on one dimension.

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We often discuss variables in terms of a regression equation:

$$Y = \alpha + \beta X + \epsilon$$

You will hear people say 'left hand side' for the dependent variable (Y) and 'right hand side' for the independent variable (x).

What are (multivariate) regressions for?

$$Y = \alpha + \beta_1 X + \beta_2 Z + \epsilon$$

$$\beta_1 = E[Y|X = x + 1, Z = z] - E[Y|X = x, Z = z]$$

- Summary: Regressing Y on X and Z, the coefficient of X, β_1, is the expected difference in Y per difference in X, comparing pairs of items that differ in X but are identical in Z.

What are (multivariate) regressions for?

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$$\beta_1 = E[Y|X = x + 1, Z = z] - E[Y|X = x, Z = z]$$

- Summary: Regressing Y on X and Z, the coefficient of X, β_1, is the expected difference in Y per difference in X, comparing pairs of items that differ in X but are identical in Z.
- Prediction: Regressions give us a model to predict some continuous outcome given linear extrapolations of our data, X and Z.

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- Prediction: Regressions give us a model to predict some continuous outcome given linear extrapolations of our data, X and Z.
- Causation: Regressions give us a model to predict how the world would have been given different values of X, where Z is a potential confounding factor associated with X and Y.

Instructions going forward

- Work through Pemberton & Rau Ch. 5
- Finish homework.